

Thermodynamics of records gives a trade-off and no floor

Result, 2026-09-27 (a recorded failure of one route). Theorem I of the unit-and-indeterminacy note says that back-action indeterminacy is impossible for classical models with Liouville dynamics, product preparations obeying a prior density ceiling, and Bayesian conditioning on records. A natural attempt to escape it adds a heat bath and requires records to be physical: stored, and eventually erased, at temperature $T > 0$. This note shows that the attempt yields an exponential trade-off between dissipated energy and posterior area, and no positive floor:

$$\eta \geq A_0 e^{-W/(k_B T)},$$

where A_0 is the prior phase-space area of the body, η the area of its posterior support after the records, and W the total work spent on measurement and erasure (Theorem 1). For every $\eta > 0$ the construction of Theorem I reaches η with finitely many finite-precision readings, hence finite W . Thermodynamics therefore prices resolution logarithmically, $W \geq k_B T \ln(A_0/\eta)$, and cannot single out an action. The would-be unit $k_B T \tau$ built from the bath and a record time rescales under the similarity of Theorem B of the dimensional note. So the thermodynamic route supplies neither ingredient (U) nor (I), and the question of STATE item 2 stays with the three denials listed in the unit-and-indeterminacy note.

The ingredients are standard (Landauer, Bennett, the Sagawa–Ueda bound for measurement plus erasure); the note records their consequence for this programme, with no novelty claimed.

1. The bound

Setting: the classical model of Theorem I, with body and pointers coupled to a heat bath at temperature T . Records are pointer states that are read, used, and eventually reset to a standard state. The generalized second law for measurement and erasure (classical form of the bound of Sagawa and Ueda 2009, metadata; the erasure part is Landauer 1961 and Bennett 1982, metadata) states that the total work of a measurement that gains mutual information I (in nats) about the system, followed by erasure of the record, satisfies $W_{\text{meas}} + W_{\text{eras}} \geq k_B T I$.

Theorem 1. Let the body’s prior be uniform on a phase-space region of area A_0 , and let a protocol of records leave, for almost every outcome, a body posterior supported in a region of area at most η . Then the total work of measurement and erasure satisfies

$$W \geq k_B T \ln \frac{A_0}{\eta}, \quad \text{equivalently} \quad \eta \geq A_0 e^{-W/(k_B T)}.$$

Proof. The differential entropy of the uniform prior is $\ln A_0$ (area measured in any fixed unit), and a density supported in area at most η has differential entropy at most $\ln \eta$. The mutual information between body and records is the prior entropy minus the average posterior entropy, so $I \geq \ln(A_0/\eta)$; the unit of area cancels. Apply the measurement-plus-erasure bound. $\square$

The construction of Theorem I is thermodynamically cheap. Its two pointers are read to precisions ϵ' and ϵ'_2 ; each reading stores finitely many bits for a bounded pointer range, and erasing them costs a finite multiple of $k_B T \ln 2$. So every posterior area $\eta > 0$ is reached at finite work, and the bound of Theorem 1 is the only thermodynamic constraint.

2. What this says about the necessity question

- **No floor.** The minimal posterior area at work budget W is $A_0 e^{-W/(k_B T)}$, which tends to zero as the budget grows. A record of Newton’s sagitta at resolution s with impulse resolution δp costs at least $k_B T \ln(A_0/(s \delta p))$: resolving a phase-space area a thousand

times below $\hbar$ costs only $k_B T \ln 1000$ more than resolving $\hbar$ itself. Thermodynamics cannot distinguish $\hbar$.

- **No unit.** The constants of the route are $k_B T$ and whatever times the protocol uses. An action $k_B T \tau$ rescales with τ , and the admitted class is closed under slowing the protocol, so Theorem B of the dimensional note gives floor zero. The universal unit of Theorem U enters through the equilibrium spectrum of radiation, which the record bound never uses.
- **The exponential again.** The trade-off is a Boltzmann factor, as in the exponential ladder of the dimension-ladder note, §4. Here the exponent contains no action constant, so the factor suppresses without setting a scale; in Wien's tail $e^{-h\nu/(k_B T)}$ the same factor sets one, because h appears in the exponent.

3. Consequence for STATE

The thermodynamic route to (I) is closed at theorem level: records with Landauer costs satisfy Theorem I with an exponential work–area trade-off. STATE item 2 keeps its three candidate denials (of Liouville dynamics, of product preparations, of Bayesian conditioning); the background and inflexion routes are under evaluation in the 2026-09-27 Astra run.